

Question Paper

Exam Date & Time: 17-Jun-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. EXAMINATIONS - JUNE 2025
SUBJECT: MIE 1071/MIE_1071 - BASIC MECHANICAL ENGINEERING
(MAKE-UP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) A boiler produces 1200 kg of steam per hour at a pressure of 12 bar and a dryness fraction of 0.85. (4)
This steam is then passed through a superheater, where its degree of super heat is 150°C above the saturation temperature. The feedwater supplied to the boiler is at a temperature of 40°C .
Determine:
a) The total amount of heat added per hour in the boiler to convert the feedwater into wet steam.
b) The total heat energy absorbed per hour in the superheater to raise the temperature of the steam.
(Assume $C_p = 4.18 \text{ kJ/kgK}$ & $C_{sup} = 2.25 \text{ kJ/kgK}$)

Course Outcome: "CO3" / Cognitive Level: "Remember"

- 1B) A boiler installed in a food processing unit generates 0.85 tons of steam per day at a pressure of 5 bar with a dryness fraction of 0.9. The boiler is equipped with an economizer, which preheats the feedwater to 85°C before it enters the boiler. It is fired using a mixture of biogas and sugarcane bagasse in equal proportions by mass. Calorific value (GCV) of Biogas = $19,500 \text{ kJ/kg}$ And Sugarcane bagasse = $8,500 \text{ kJ/kg}$. If the boiler operates with an average thermal efficiency of 80%, calculate the daily consumption of biogas and sugarcane bagasse. Assume specific heat of water and superheated steam as 4.187 kJ/kg K and 2.25 kJ/kg K respectively. (3)

Course Outcome: "CO3" / Cognitive Level: "Remember"

- 1C) Compare the functions of an economizer and an air preheater. How do they complement each other in a boiler system? (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2A) An open belt drive 10 cm wide connects two pulleys mounted on parallel shafts 2.5m apart. The diameter of driving pulley is 45cm and that of follower is 30 cm. If permissible pull in the belt is limited to 180 N/cm of width, find the power that can be transmitted when driving pulley is rotating at 120 rpm also calculate the length of the belt. Assume co-efficient of friction is 0.3. (4)

Course Outcome: "CO3" / Cognitive Level: "Remember"

- 2B) The shaft from a prime mover consists of six gear wheels A, B, C, D, E & F. Gears B and C are compound gears, as are gears D and E. Gear E meshes with gear F and gear A meshes with gear B which is mounted on the driven shaft. The number of teeth on gears A, B, C, D, E, and F are 30, 45, 60, 72, 84, and 96 respectively. Gear A which rotates at 200rpm in anticlockwise direction. Sketch the arrangement and determine the speed of driven gear and calculate the centre distance (3)

between the driver and driven shafts if the module of gears is 2π mm.

Course Outcome: "CO3" / Cognitive Level: "Remember"

- 2C) Explain the function of fast and loose pulley with a neat sketch. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) With suitable illustrations explain how the reciprocating motion is transformed into the rotary motion in a prime mover that indirectly admits the charge and operates on the principle of isobaric heat addition? (4)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3B) The following data refers to a twin cylinder I C Engine: (3)
Cylinder diameter = 150mm, Stroke length = 200mm, Total cam shaft revolutions/hour = 12000,
Mean effective pressure = 0.64MPa, Fuel consumption = 10litres/hour, Calorific value of the fuel = 45000kJ/kg, Density of the fuel = 780kg/m³, Determine the indicated thermal efficiency.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 3C) How do two stroke petrol engines differ from two stroke diesel engines? List any six points. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4A) Suggest a suitable lathe operation which is used to produce helical grooves on a cylindrical workpiece. Explain the same with the help of a neat sketch. (4)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 4B) Suggest and explain, with the help of a neat sketch, suitable drilling operations that can be performed to accomplish the following: (3)
i) Flat seating surface for a washer
ii) Groove for a bolt head

Course Outcome: "CO2" / Cognitive Level: "Application"

- 4C) Sketch and explain different types of gas flames that can be used in oxy-acetylene welding process. (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 5A) Explain the significance of the following pattern allowances: (4)
i) Machining allowance
ii) Shrinkage allowance
iii) Rapping allowance
iv) Draft allowance

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 5B) List down any six reasons for employing automation in production. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 5C) How do the following components of an NC machine contribute to making the system unreliable: Punched tape, control unit and tape reader. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

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Event Name	Subject
MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025	MIE 1071-PHY:BASIC MECHANICAL ENGINEERING

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	16
3	3	CO 3	31
Total			50

* **NA** - CO Not Selected

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#	Level	No of Questions	Marks (%)
1	Remember	4	14(28%)
2	Understand	7	23(46%)
3	Application	4	13(26%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)